SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA1610 COURSE CODE: Data Warehousing and Data Mining

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COURSE OUTCOME COVERED

CO3: Evaluate the effectiveness of classification techniques on real-time data for accurate prediction, enabling informed decision-making. (BL3,BL4,BL5)
Assessment Tool Weightage: 30%

QUESTIONS: 3 **Total Marks:30**

Annexure C

Case study

Criteria	Understanding of Case *(2.5)*	Application of Concepts *(2.5)*	Analysis *(2)*	Solution Design *(2)*	Clarity *(1)*	*Total(10)*
Weightage	*25%*	25%	20%	20%	10%	*100%*
Q1						
Q2						
Q3						

Code of Conduct:

I G.Hima Bindu(192511377) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: G.Hima Bindu

ANNEXURE A

Case Study:

Q1: Shepherd Boy – Wolf Case (Confusion Matrix)

- **Scenario:**
  - Boy cries "Wolf!" when there is no wolf → villagers respond incorrectly.
  - Boy cries "Wolf!" when there is a wolf → villagers ignore him.
- **Mapping to Confusion Matrix:**
  - **True Positive (TP):** Wolf present, villagers respond → *Never happens here*.
  - **False Positive (FP):** Wolf absent, villagers respond → *Villagers fooled by boy's lie*.
  - **False Negative (FN):** Wolf present, villagers don't respond → *Villagers ignore real wolf*.
  - **True Negative (TN):** Wolf absent, villagers don't respond → *Never happens here*.

👉 **Answer:**

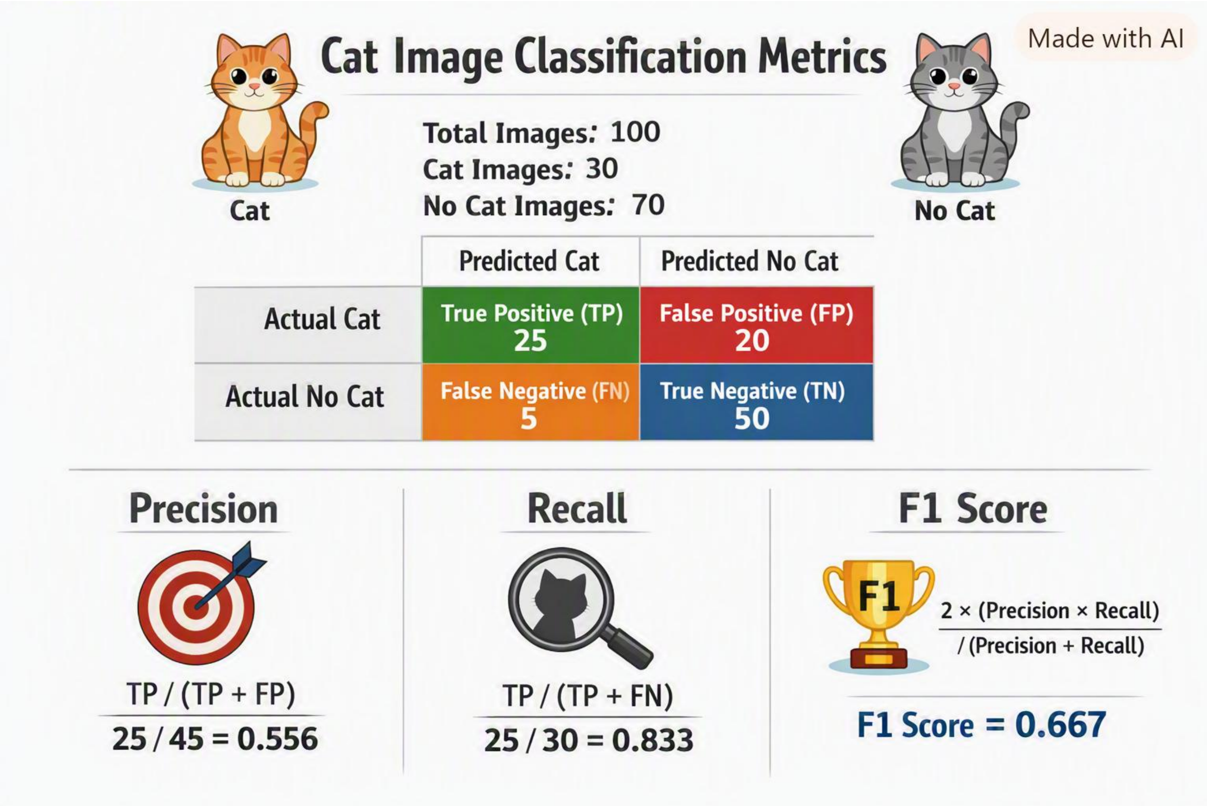
- TP = 0
- FP = 1 (villagers respond to false alarm)
- FN = 1 (villagers ignore real wolf)
- TN = 0

Q2: Cat Image Classification

- **Dataset:**
  - Total images = 100
  - Cat images = 30
  - Non-cat images = 70
- **Model Predictions:**
  - Correctly predicts cat in 25/30 → **TP = 25**
  - Misses 5 cat images → **FN = 5**
  - Correctly predicts no cat in 50/70 → **TN = 50**
  - Wrongly predicts cat in 20 non-cat images → **FP = 20**
- **Metrics:**
  - **Precision = TP / (TP + FP) = 25 / (25 + 20) = 25 / 45 ≈ 0.556**
  - **Recall = TP / (TP + FN) = 25 / (25 + 5) = 25 / 30 ≈ 0.833**
  - **F1 Score = 2 × (Precision × Recall) / (Precision + Recall)** = 2 × (0.556 × 0.833) / (0.556 + 0.833) = 2 × 0.463 / 1.389 ≈ **0.667**

👉 **Answer:**

- TP = 25, FP = 20, FN = 5, TN = 50
- Precision ≈ 0.556
- Recall ≈ 0.833
- F1 Score ≈ 0.667



Q3: Email Spam Classification (Naïve Bayes)

Step 1: Prior Probabilities

- Total samples = 6
- Spam = 3 → **P(Spam) = 3/6 = 0.5**
- Not Spam = 3 → **P(Not Spam) = 3/6 = 0.5**

Step 2: Conditional Probabilities

- For **Spam (3 samples):**
  - Offer=Yes → 2/3
  - Offer=No → 1/3
  - Free=Yes → 2/3
  - Free=No → 1/3
  - Length=Short → 2/3
  - Length=Long → 1/3
- For **Not Spam (3 samples):**
  - Offer=Yes → 1/3
  - Offer=No → 2/3
  - Free=Yes → 0/3 = 0
  - Free=No → 3/3 = 1
  - Length=Short → 1/3
  - Length=Long → 2/3

Step 3: Classify (Offer=Yes, Free=No, Length=Short)

- **For Spam:** P(Spam) × P(Offer=Yes|Spam) × P(Free=No|Spam) × P(Length=Short|Spam) = 0.5 × (2/3) × (1/3) × (2/3) = 0.5 × 4/27 ≈ **0.074**
- **For Not Spam:** P(Not Spam) × P(Offer=Yes|Not Spam) × P(Free=No|Not Spam) × P(Length=Short|Not Spam) = 0.5 × (1/3) × (1) × (1/3) = 0.5 × 1/9 ≈ **0.056**

👉 **Prediction:** Since 0.074 > 0.056 → **Class = Spam**

Step 4: Compare with WEKA

- If you run this dataset in WEKA with Naïve Bayes, you'll get the same classification result: **Spam**.

Step 5: Independence Assumption

- Naïve Bayes assumes features are **independent given the class**.
- In reality, features like "Offer" and "Free" may be correlated (spam emails often contain both).
- Despite this unrealistic assumption, Naïve Bayes often performs well because it focuses on probability ratios.

🎯 **Final Structured Answers**

- **Q1:** TP=0, FP=1, FN=1, TN=0
- **Q2:** TP=25, FP=20, FN=5, TN=50; Precision=0.556, Recall=0.833, F1=0.667
- **Q3:**
  - P(Spam)=0.5, P(Not Spam)=0.5
  - Conditional probabilities computed above
  - Classification result: **Spam**
  - WEKA confirms same result
  - Independence assumption: unrealistic but effective in practice


Made with AI
Naive Bayes
Spam Classification
SPAM
Sample Email:
Offer = Yes, Free = No,
Length = Short
Offer
Free?
Short
Email
Calculate Probabilities:
P(Spam)
P(Not Spam)
P(Spam) × P(Offer=Yes | Spam) ×
P(Free=No | Spam) ×
P(Short | Spam)
$= 0.5 \times \frac{2}{2} \times \frac{1}{3} \approx 0.074$
P(Not Spam) × P(Offer=Yes | Not Spam)
P(Free=No | Not Spam) ×
P(Short | Not Spam)
$= 0.5 \times \frac{1}{3} \times \frac{1}{1} \approx 0.056$
Compare Probabilities:
0.074 vs 0.056
Result:
Classified as: SPAM
SPAM

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Case	25	Thorough understanding; clearly identifies all key issues	Good understanding with most issues identified	Basic understanding; some issues missed	Poor understanding; problem not identified correctly
Application of Concepts	25	Effectively applies relevant concepts to analyze the case deeply	Applies appropriate concepts with minor gaps	Limited application of concepts	Fails to apply relevant concepts
Analysis	20	In-depth analysis with strong reasoning and insights	Good analysis with logical reasoning	Basic analysis with limited depth	Weak or no analysis; lacks reasoning
Solution Design	20	Innovative, feasible solutions with strong justification	Logical solutions with adequate justification	Basic solutions with weak justification	Incorrect or no solution provided
Clarity	10	Clear, well-structured, and easy to understand	Organized and understandable presentation	Limited clarity and structure	Poor clarity; difficult to understand